Ian Nelson

509.212.7650 | ianinafrica9@gmail.com | linkedin.com/in/ian-nelson-95b767352 | iannelson9.github.io/portfolio

Education

Liberty University May 2028 (anticipated)

BS: Mechanical Engineering, GPA 3.75/4.0

Technical Skills

Software & Methods: SOLIDWORKS (CAD, PDM, Simulation, FEA), Ansys, Excel, DFM, GD&T

Manufacturing: Manual Lathe, Bandsaw, Drill Press Operation

Experience

Formula SAE – Vehicle Dynamics Engineer *Fall 2024 – Present*

- Designs suspension and steering geometry to meet vehicle dynamics performance targets.
- Performs design for manufacturing (DFM) reviews on complex parts to allow for in-house manufacturing.
- Achieved a ~50% weight reduction in suspension hardware, minimizing unsprung mass and enhancing tire-to-road contact.
- Developing lap time simulation using Python to evaluate suspension sensitivity.

Valmet – Mechanical Engineering Intern *Summer 2025*

- Created CAD models and drawings for tube shield die manufacturing using GD&T specifications.
- Designed sizing tool to enable manufacturing of odd sizes for tube shield dies.
- Improved Excel tools for key products, enabling item number lookup within pricing spreadsheets.

STEM Outreach & Community Service – Ghana, West Africa *2015-2024*

- Tutored African middle schoolers in math and science.
- Helped lead educational events and youth activities.